

EFFECT OF TIMING OF THE MODEL TEXT ON ALIGNMENT IN THE ITERATIVE CONTINUATION TRANSLATION TASK

JADE JIAXUAN DANG^{1,2}, MIN WANG²

¹School of Philosophy, Psychology & Language Sciences, University of Edinburgh;
²School of Foreign Studies, Xi'an Jiaotong University

BACKGROUND

When second language (L2) learners mirror words from a bilingual model text during translating, it may lead to increased proficiency. Previous research has measured the effect of alignment by counting repetitions or manual annotation of decontextualized words (especially content words), expressions, sentence structures, and translation skills.

RESEARCH QUESTIONS

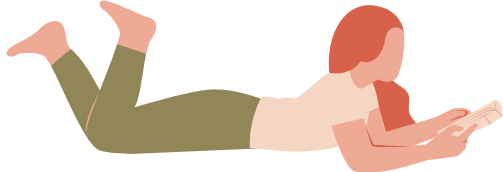
1. Does alignment occur?

To what degree L2 learners match the content or style of their output with the input, making it possible to observe alignment beyond traditional linguistic features, that is, the repetition of exact words.



2. How to measure?

In the translation field, even less is known about whether function words are reused appropriately, whether content words are used creatively in new contexts thereby leading to the increase in overall translating ability.



3. To what extent?

To what extent does the timing of the model text modulate the alignment effect in the continuation translation task?

METHODS

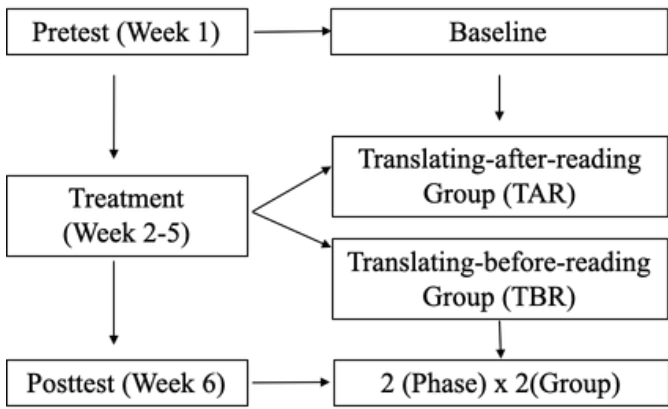
PARTICIPANTS

40+ Chinese undergraduates
(Major in English Translation)

MATERIALS

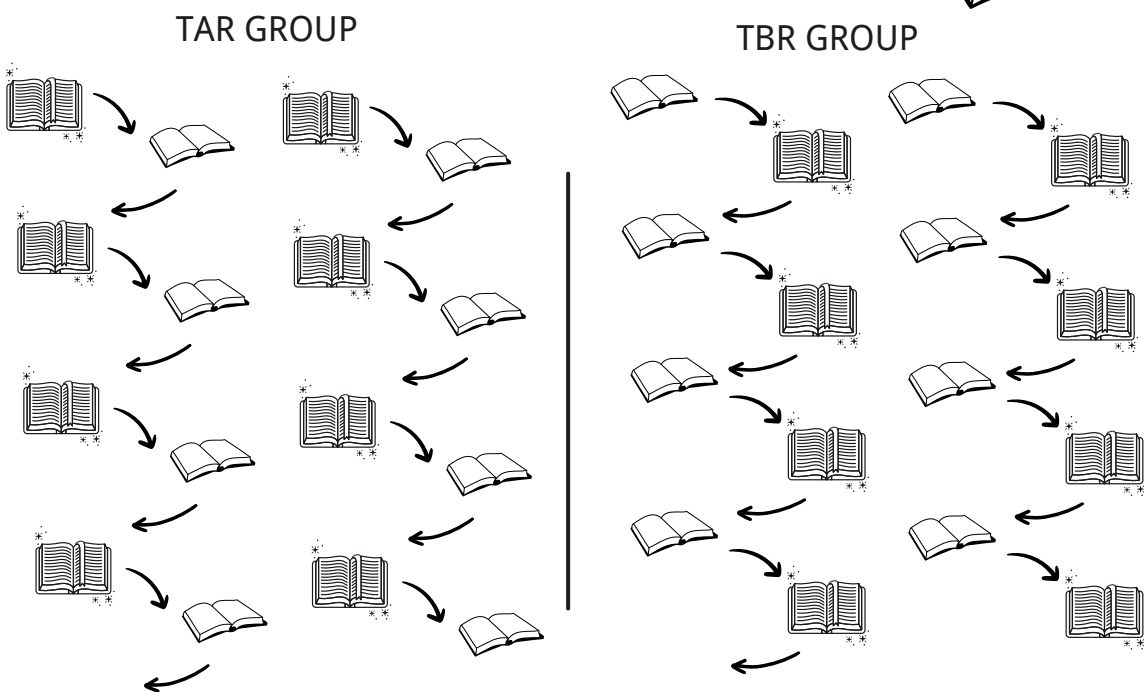
bilingual prose text
朱自清(C) + Howard Goldblatt(E)

RESEARCH DESIGN



TASK

Continuation translation task



MEASURES

- Linguistic Style Matching (LSM)

LSM is analogous to factor analysis of **function words** based on the relationship between pairs of documents using word co-occurrence data. It enables the measurement of semantic similarity among non-identical words.

- Quantitative Index Text Analyser (QUITA)

Examples:

h-point

boundary point of function and content words

$$h = \frac{f(i)j - f(j)i}{j - i + f(i) - f(j)}$$

content concreteness

non-abstract of words

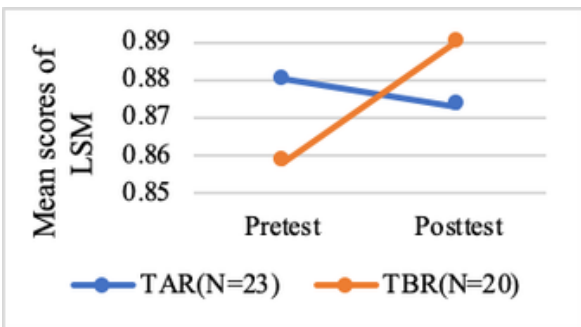
$$Conc(D) = \frac{1}{|D|} \sum_{t \in D} Conc(t)$$



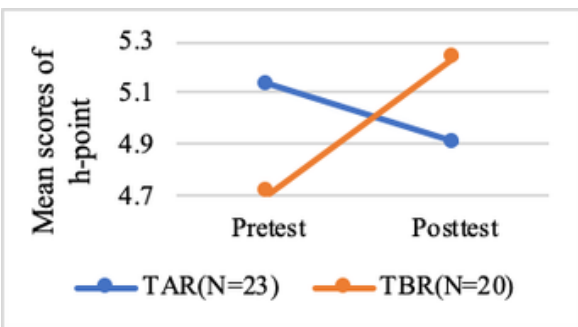
INTENDED RESULTS

- The TBR group evoked stronger conceptual and lexical alignment of function words.
- Both LSM and h-point improved from the pretest to the posttest.
- The mean LSM score for **the TBR group should be higher** than that for the TAR group during the treatment phase.

MOCK DATA FIGURES



(a) LSM



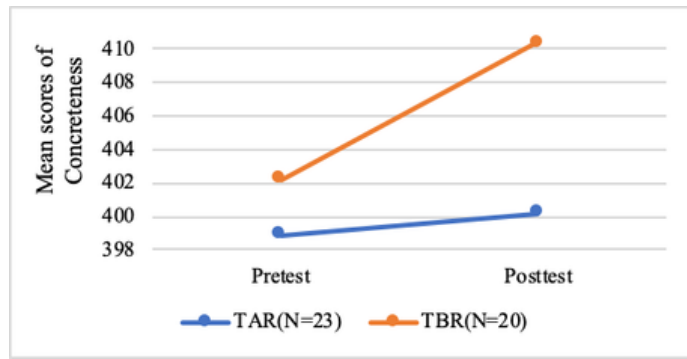
(b) h-point

Function index of different groups between pre/post phases: Mean (SD)



- The mean score of the TBR group rose sharper than that of the TAR group between the pretest and the posttest.
- There should be **an interaction between phase and group**.

MOCK DATA FIGURES



Content index of different groups between pre/post phases: Mean (SD)

PREDICTIONS OF FINDINGS

- This study will extend the scope of inquiry of L2 alignment from content words to **function words**. It is thus demonstrated that the alignment effect may also be manifested in L2 learners' mimicry of the **language style** of the input.
- Furthermore, the timing of the model text may show a **dual effect**, both as **corrective feedback** to the previous translation task and as **priming** towards the following task, despite the time lag in the latter. More specific examples from the corpus and further analysis of the treatment period are needed to conduct.
- The dual mechanisms hopefully superpose to the TBR group incurred a deeper strength of alignment effect, thereby shedding new light on the interplay between alignment effect and L2 learners' social and psychological states.

REFERENCES

- Jiang, X., Jiang, Y. & Hol, C. (2022). Is Queen's English drifting towards common people's English?—Quantifying diachronic changes of Queen's Christmas Messages (1952-2018) with reference to BNC. *Journal of Quantitative Linguistics*, 29(1): 1-36.
- Kubát, M., Matlach, V. & Eech, R. (2014). QUITA. Quantitative Index Text Analyzer, Lüdenscheid: RAM-Verlag.
- Landauer, T., & Dumais, S. (1997). A solution to Plato's problem: The latent semantic analysis theory of acquisition, induction, and representation of knowledge. *Psychological Review*, 104(2): 211-240.
- Michel, M., & Cappellini, M. (2019). Alignment during synchronous video versus written chat L2 interactions: A methodological exploration. *Annual Review of Applied Linguistics*, 39, 189-216.
- Wang, C., & Wang, M. (2015). Effect of Alignment on L2 Written Production. *Applied Linguistics*, 36(5), 503-526.
- Zhang, S., & Zhang, L. (2021). Effects of a xu-argument based iterative continuation task on an EFL learner's linguistic and affective development: Evidence from errors, self-initiated error corrections, and foreign language learning attitude. *System*, 98.